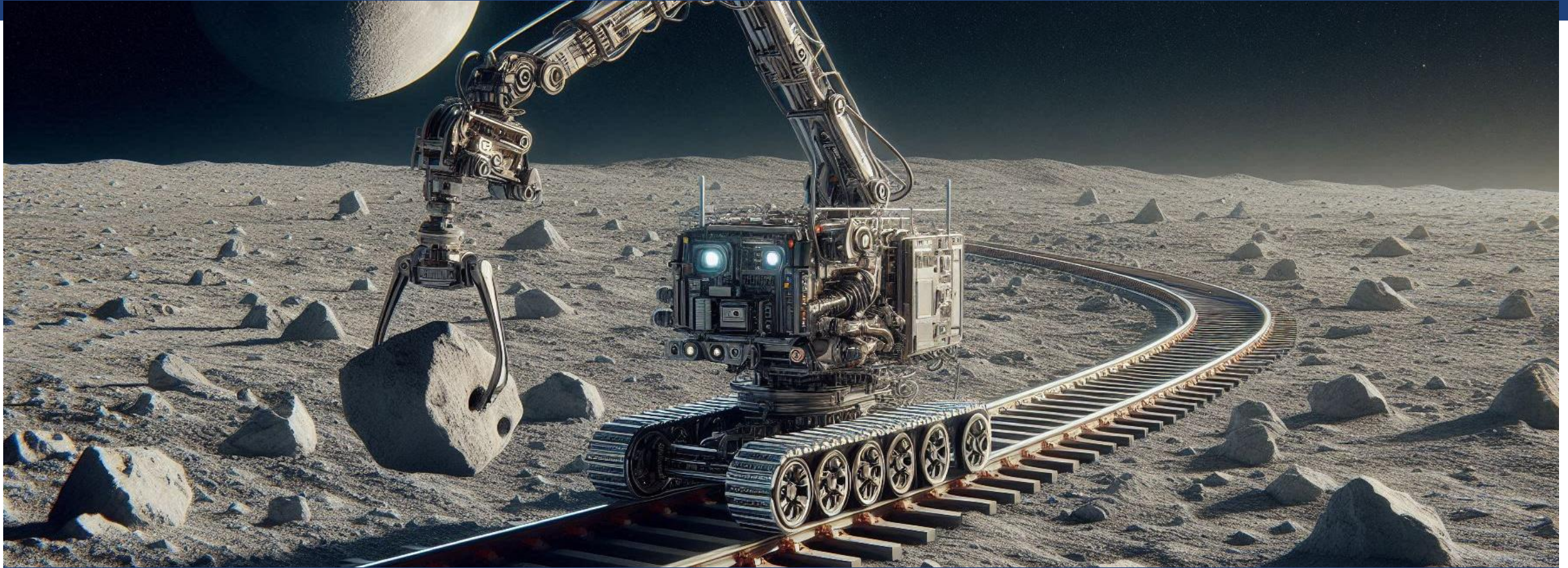




Lunar Resource Sorting Facility

System Engineering Exercise, Day 1

Dominic Ziegler, Andrina Keel

TA Contacts

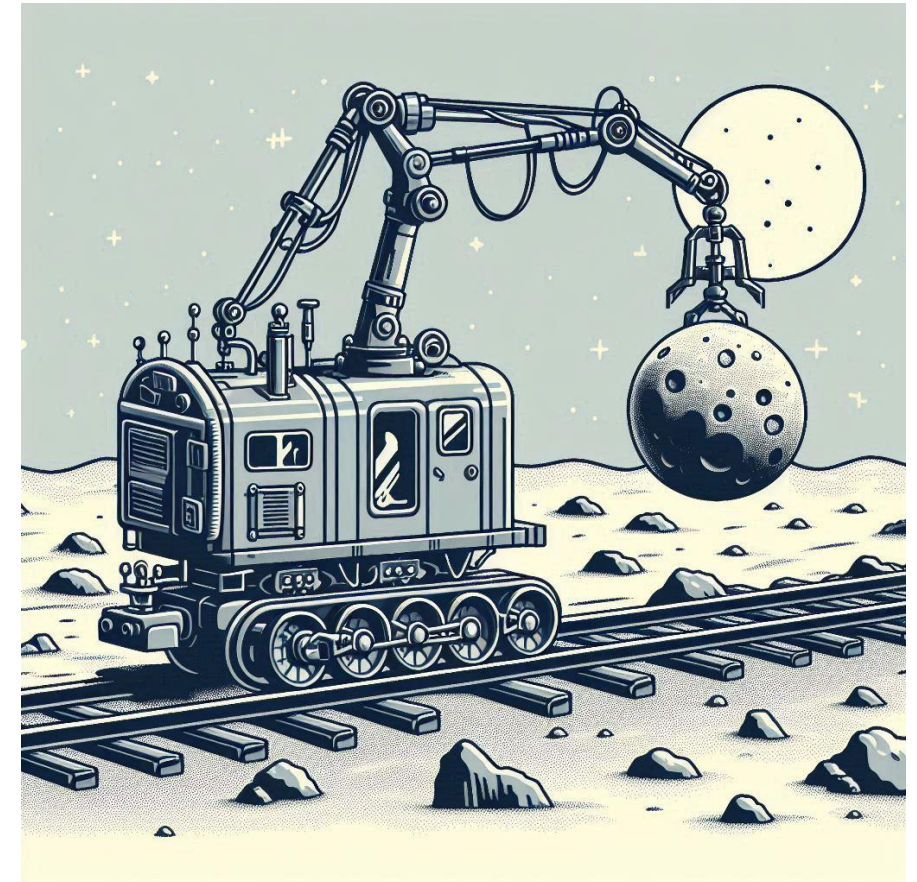
- Andrina Keel ankeel@student.ethz.ch
- Dominic Ziegler doziegler@student.ethz.ch

Welcome to ETH Hangar Dübendorf !

- 24/7 access with Legi and PIN until October 5th
- Available Workspace in E10 and on 2nd floor

Storyline

- Field of Resources discovered on the Moon
- Resources are vital for crewed lunar base
- Space agency issues call for proposals for resource collection robot design competition
- Create an autonomous robot to...
 - Identify resources
 - Collect resources
 - Transport resources
 - Sort resources



Goal of this Exercise

- Teambuilding
- Intro to System Engineering Workflow & Practices
-> intentionally ungraded

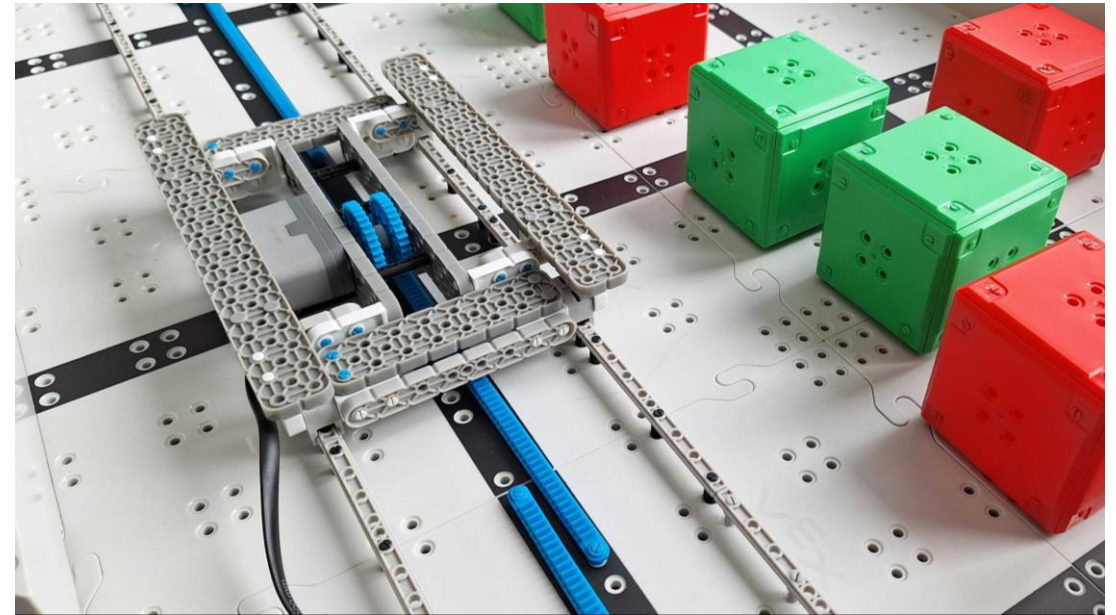
BUT

There will be prizes to win!



Task Description

- Use VEX Robotics Kit to identify, collect, transport, and sort cubes from “resource-rich area” to “production facility”
- Competition with four scored deliverables:
 - Data Packages 1 & 2
 - Design Review
 - System Demonstration
- Details on scoring & rules in Task Description
- Additional Resources: SMAD



Timeline

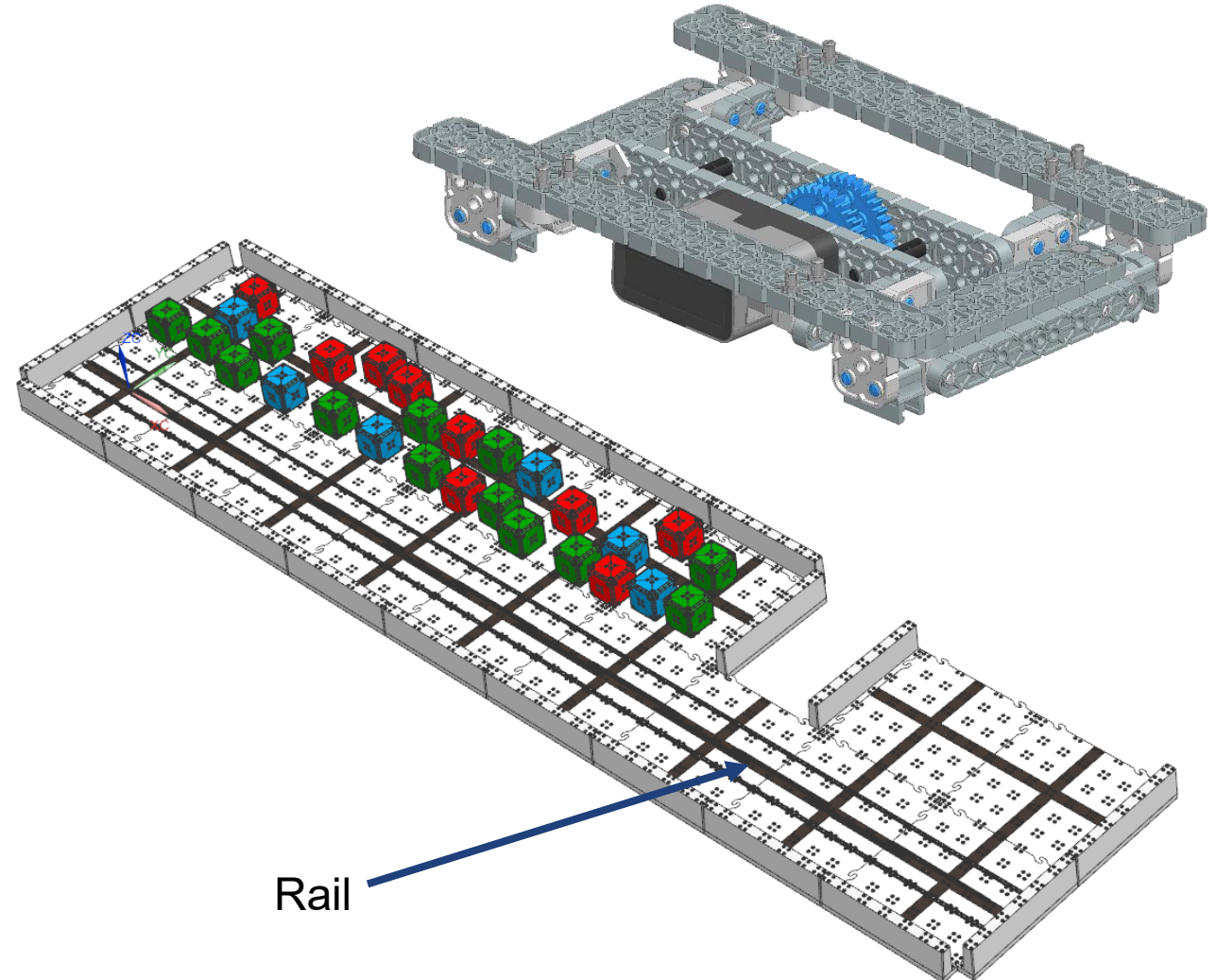


Time	Monday, 28.09. (1300-1700)		Wednesday, 30.09. (1300-1700)	Monday, 05.10. (1300-1830)
Task 1	Work on Data Package 1 - Mission Requirements - Stakeholder Analysis - Product Tree - Requirements Flow-Down		Design Review (1400-1600) - System Overview - ConOps - V-Model & FMEA - Live Demo	System Demonstration (1400-1600)
Task 2	Baseline Design (preparation for Design Review) - ConOps - V-Model & FMEA		Work on Data Package 2 - Implement Feedback - Final Requirements Flow-Down	Wrap-Up - Clean-Up - Lessons Learned & Feedback
Task 3	Baseline Design	- ConOps - Key Requirements - Design Review	Design Iteration & Testing	Announcement of Winners & Celebration
Deliverables	Data Package 1 (today by 22:59)		Design Review (by Wed 13:59) Data Package 2 (by Thu 22:59)	Final System

Lunar Resource Sorting Facility

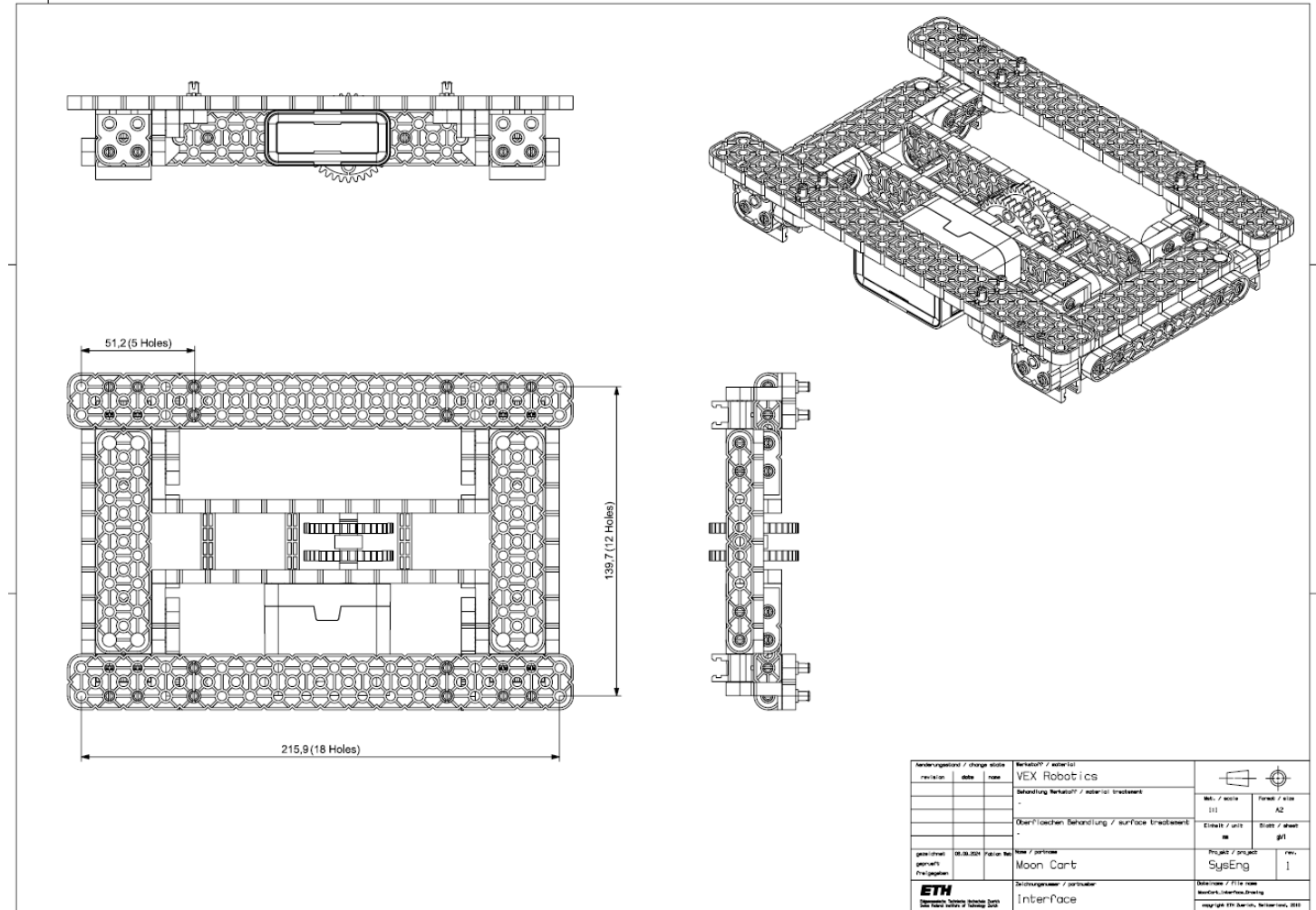
Playing Field

- Playing field equipped with rails
- Cart for interface to rails -> can be used by teams
- Resources are red & green cubes
- Blue cubes are unusable
- Cubes to be organized in “factory”



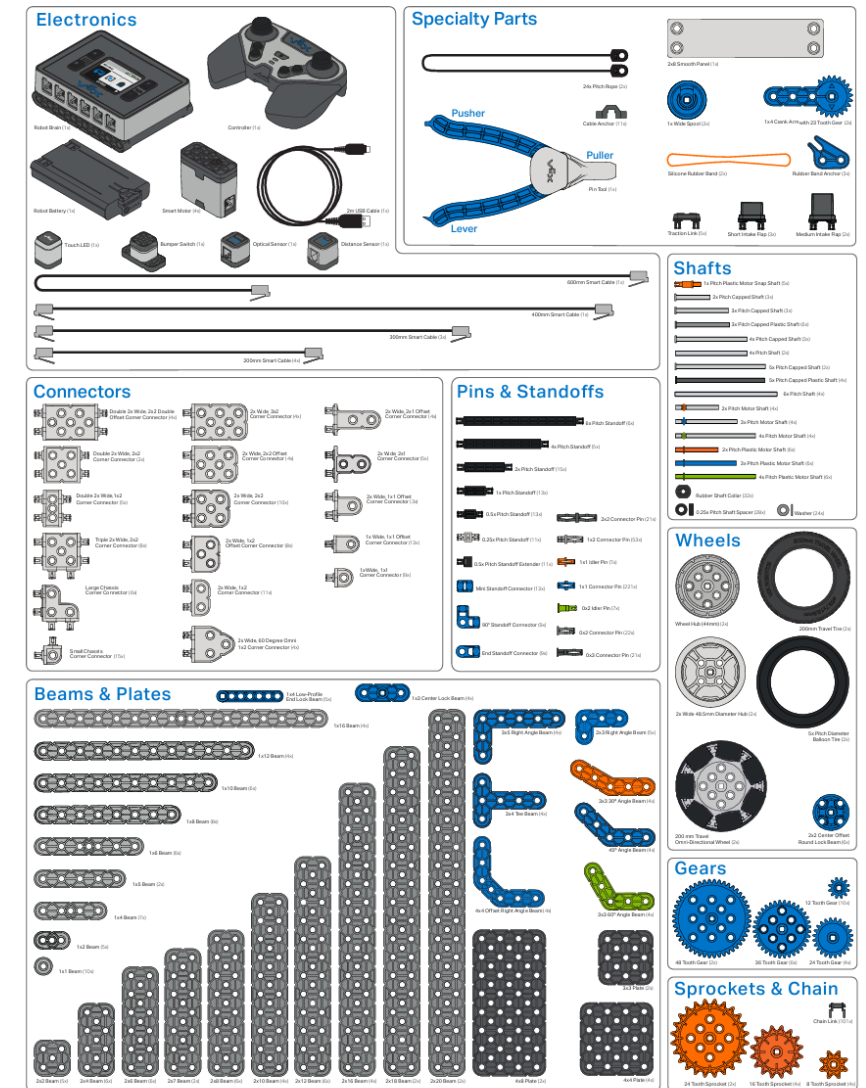
Lunar Resource Sorting Facility Cart

- Interface document provided
- In real missions usually called “ICD”: Interface Control Document
 - Mech. ICD
 - Elect. ICD
 - Therm. ICD



VEX IQ Kit

- LEGO-Style robotics system
- ~1'000 Parts & Electronic components
 - 4 Motors
 - 4 Sensors (Distance, Colour, Bumper, Capacitive Touch)
 - 1 Robot “Brain”



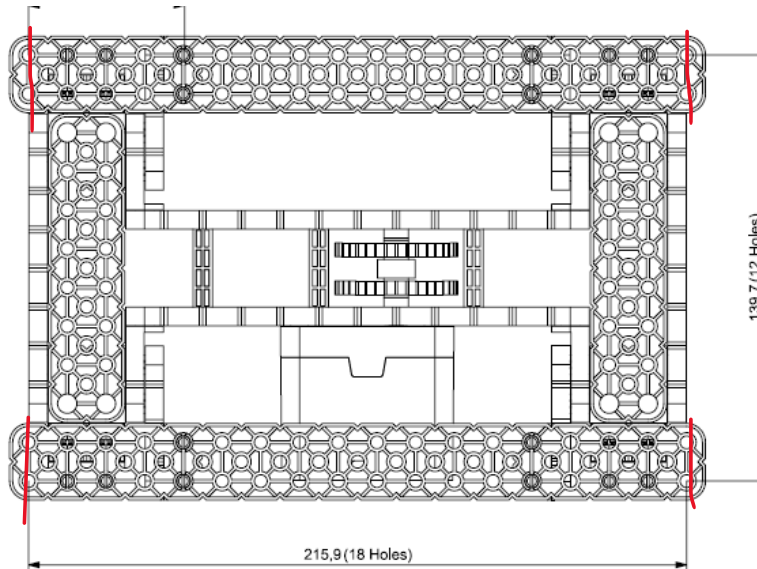
A digital version of this poster is at posters.vex.com with more helpful information and links to spare parts

VEX IQ Robot Programming

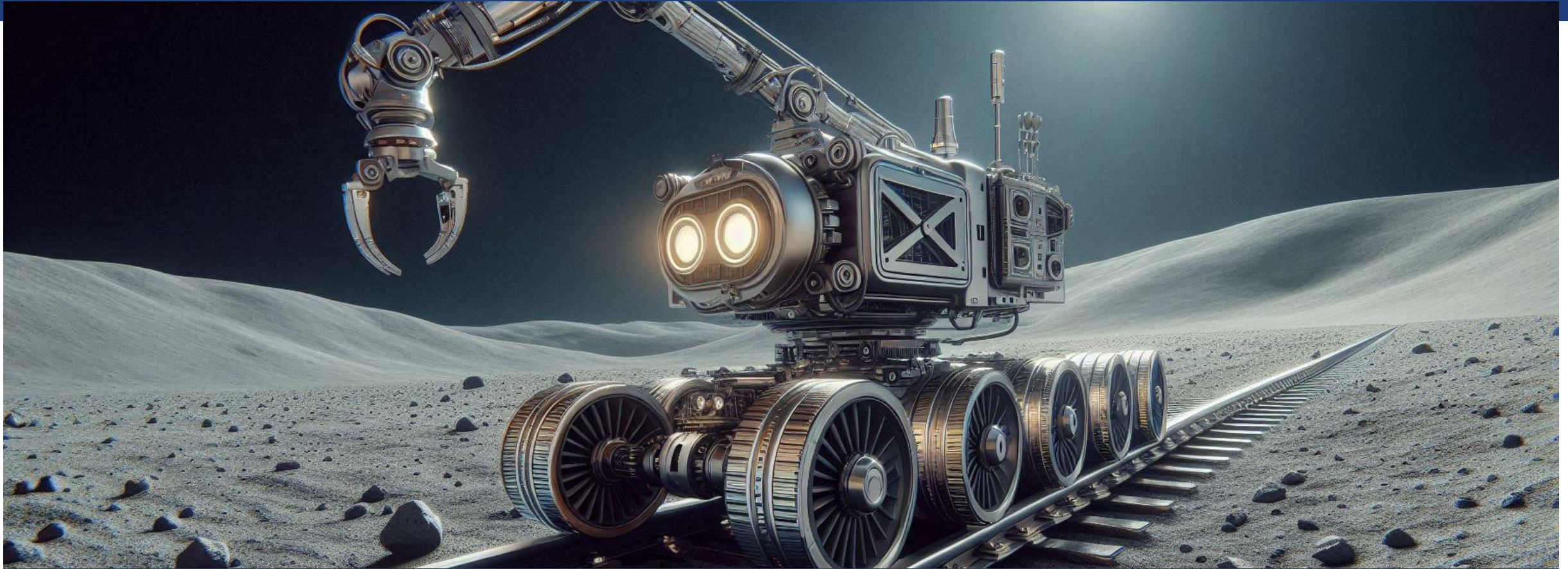
- System needs to function autonomously
- Robot Brain is programmable with Python or C++
- Check Task Description for Tips & Install instructions
- Code template and test file for all sensors and motors

Testing Slots

- 4 Test Fields available
 - 2 Teams per field
- Predefined slots of 30 mins
 - Can be organized bilaterally
- Trolley different!



Test Stand 1			
Mon. 28.09		Wed. 30.09	
Time	Team	Time	Team
		13:00	Team 1
		13:30	Team 2
14:00	Team 1	14:00	Team 1
14:30	Team 2	14:30	Team 2
15:00	Team 1	15:00	Team 1
15:30	Team 2	15:30	Team 2
16:00	Team 1	16:00	Team 1
16:30	Team 2	16:30	Team 2
17:00	Team 1	17:00	Team 1
17:30	Team 2	17:30	Team 2

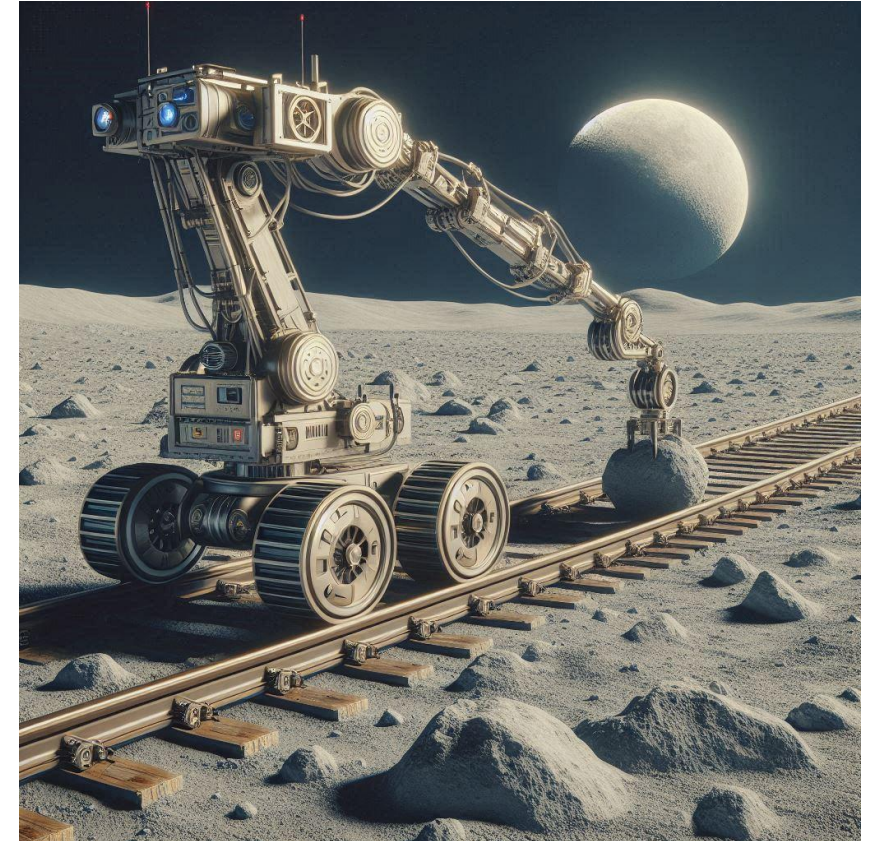
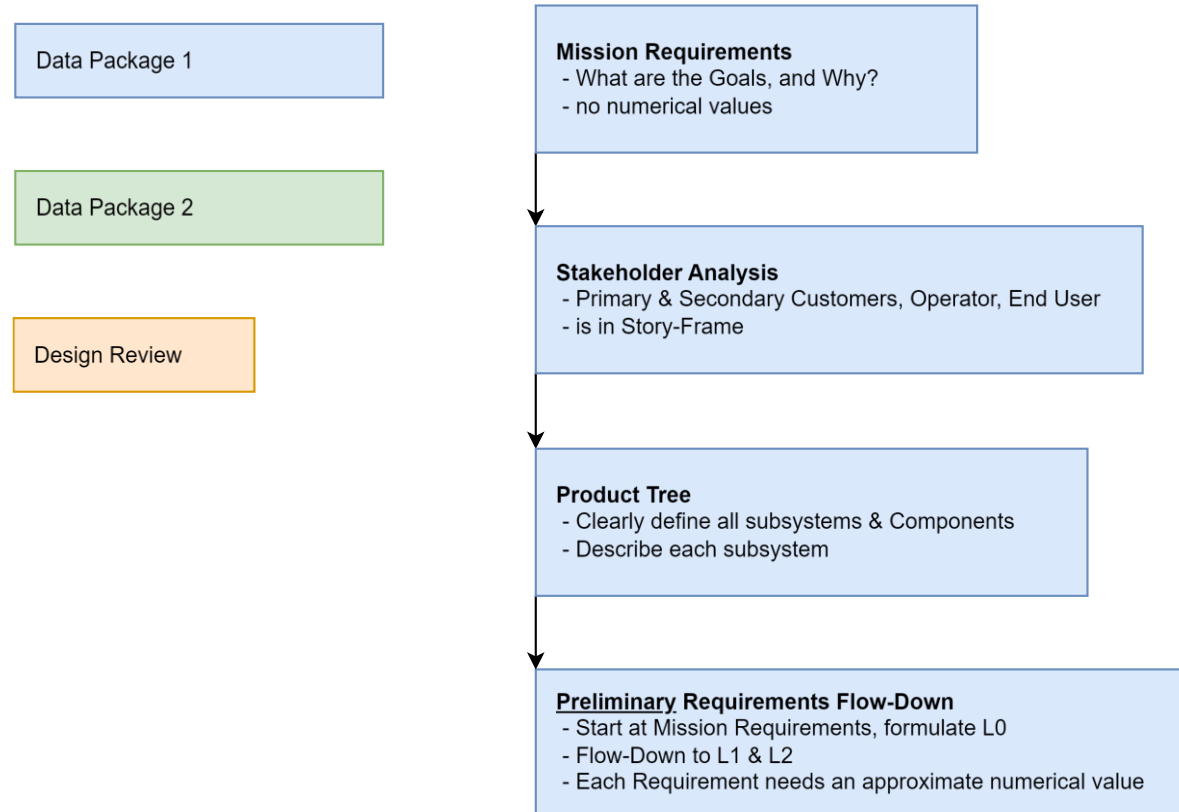


Lunar Resource Sorting Facility: Data Package 1&2

System Engineering Exercise, Day 1

Deliverables Overview, Day 1

System Engineering Exercise - Deliverables Overview



Mission Statement

- Qualitative Goal
 - “What are the goals of the mission/system and why?”
 - No quantitative statements
- Example: Kitchen cupboard



Stakeholder Analysis

- Do in “Story” frame!
- Primary Customer
- Secondary Customer
- Operator
- End User
- Template in Hand-Out

52

Space Mission Engineering

3.4

Table 3-2. The Principal Players, Their Objectives, and What Might Change Those Objectives for Representative Systems. See text for discussion.

Mission	FireSat II	SCS	Safari Airplane
Primary Customer / Objective	US Forest Service / Detect and monitor forest fires in US and Canada using as little financial resources as possible	Both DoD and Disaster Relief Agency / Supplemental, continuous, real-time communications	African safari operator / Improve profits by moving people quickly between sites
Secondary Customer / Objective	Congress / Demonstrate to the public that action is being taken and government funds spent wisely	Congress / Satisfy military needs and, at the same time, respond to public perception of slow disaster relief support	National Bank of Kenya / Make money on a commercial loan; support increased safari business
Operator / Objective	FireSat II Mission Operations Office / Provide forest fire data to the end user in real time; archive data; monitor and maintain the health and safety of FireSat II	SCS Spacecraft Operations Office / Monitor and maintain the continuously operating communications link; monitor and maintain the health and safety of the satellites	2 pilots hired by the safari company / Provide rapid, safe interlocation transportation
End User / Objective	Forest fire field management or firefighter in truck / Track current location of fire relative to themselves	Individual soldier or relief worker in the field / Explain situation and get the right help where it's needed as quickly as possible	People on safari / Don't care about airplane—just want to see as much as they can in 10 days
What Might Change System Objectives	Change in administration; change in projected system cost (up or down); major fire that didn't have sufficient monitoring resources	Splitting procurement to buy 2 systems; major disaster where communications weren't available in a timely manner	Safari company sold to either much bigger or much smaller company; a new, very remote site for wildlife viewing is opened

Product Tree

- Decomposition of System into Subsystems & Components
 - Hierarchical break-down
 - Each entry serves one function
- Additionally: Describe each subsystem
 - 1-2 sentences per subsystem
- Template in Hand-Out

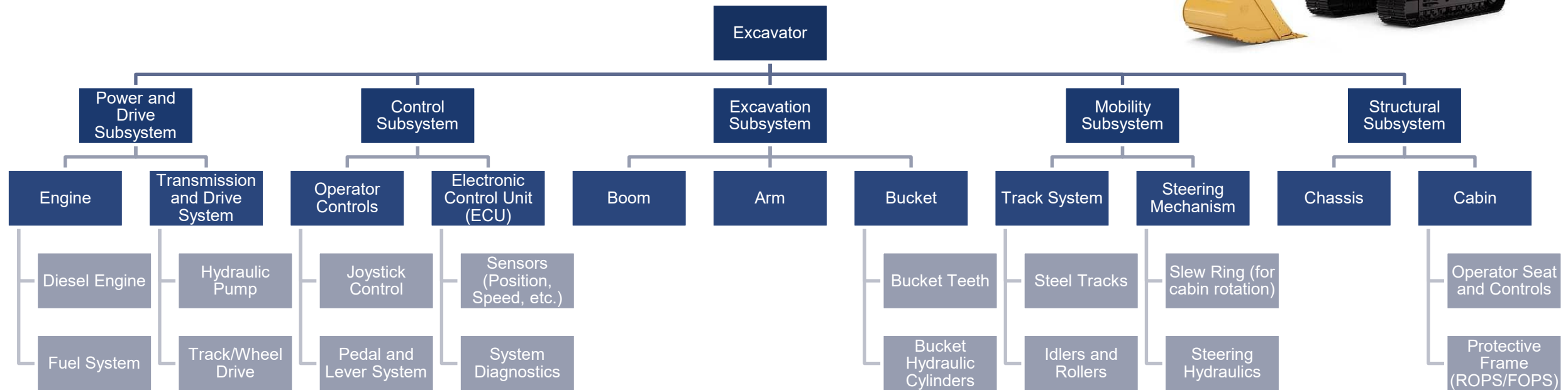
Product Tree

Incorrect Example



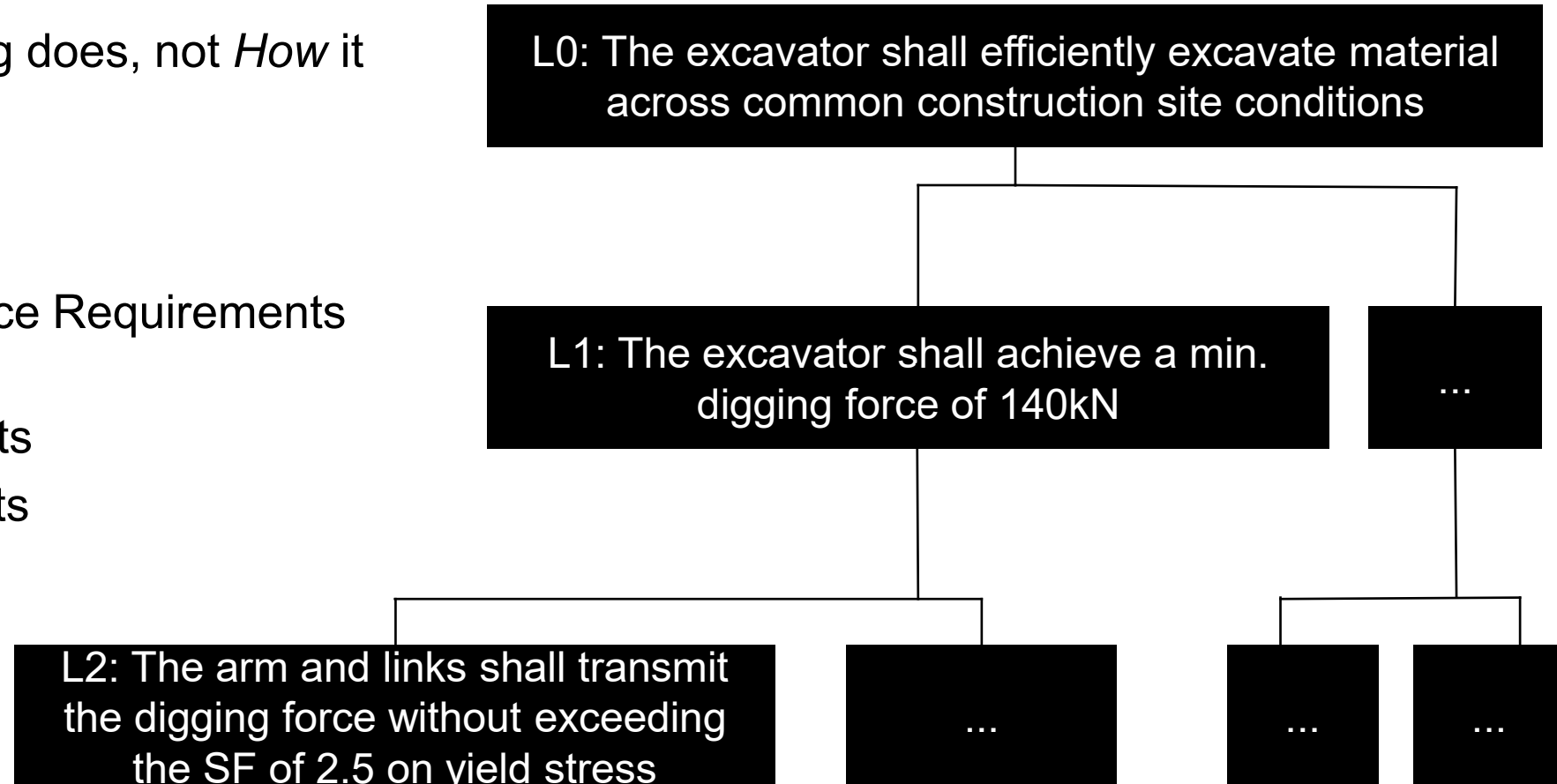
Product Tree Example

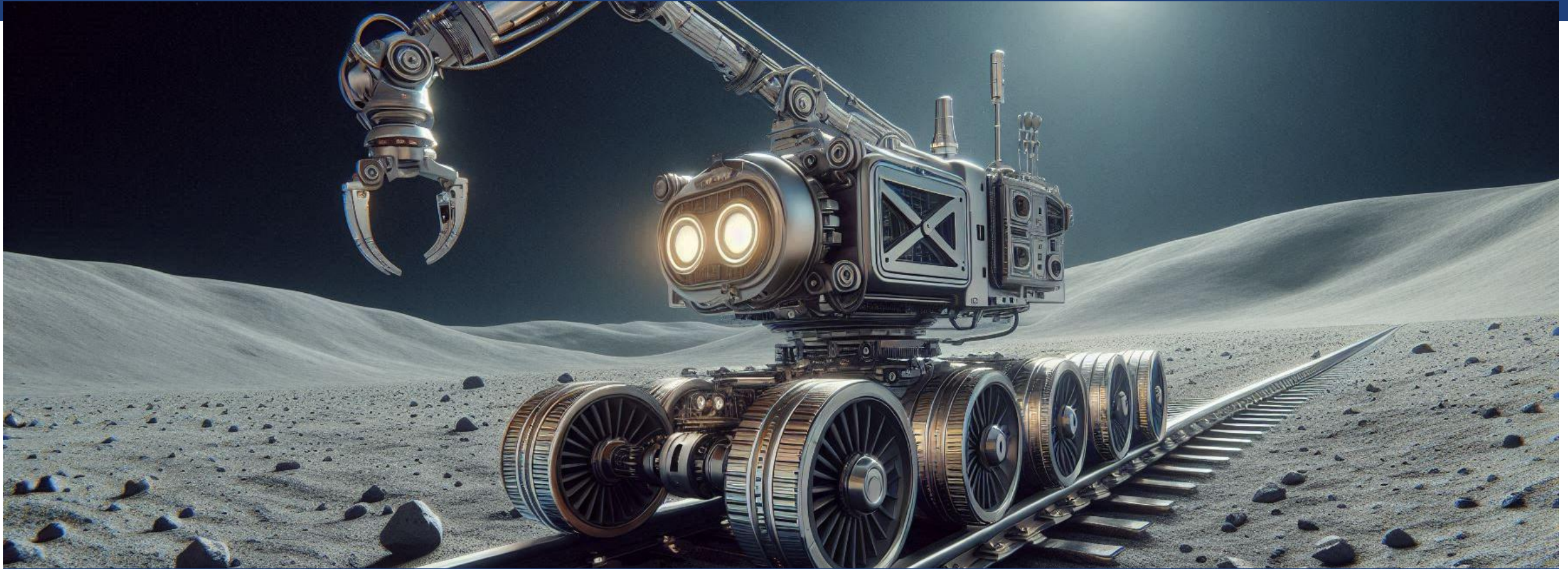
Excavator



Preliminary Requirements Flow-Down

- Describes *What* the thing does, not *How* it does it
- Traceable flowdown
- L0: Programmatic/Science Requirements and Constraints
- L1: Mission Requirements
- L2: System Requirements





Lunar Resource Sorting Facility: Design Review

System Engineering Exercise, Day 2

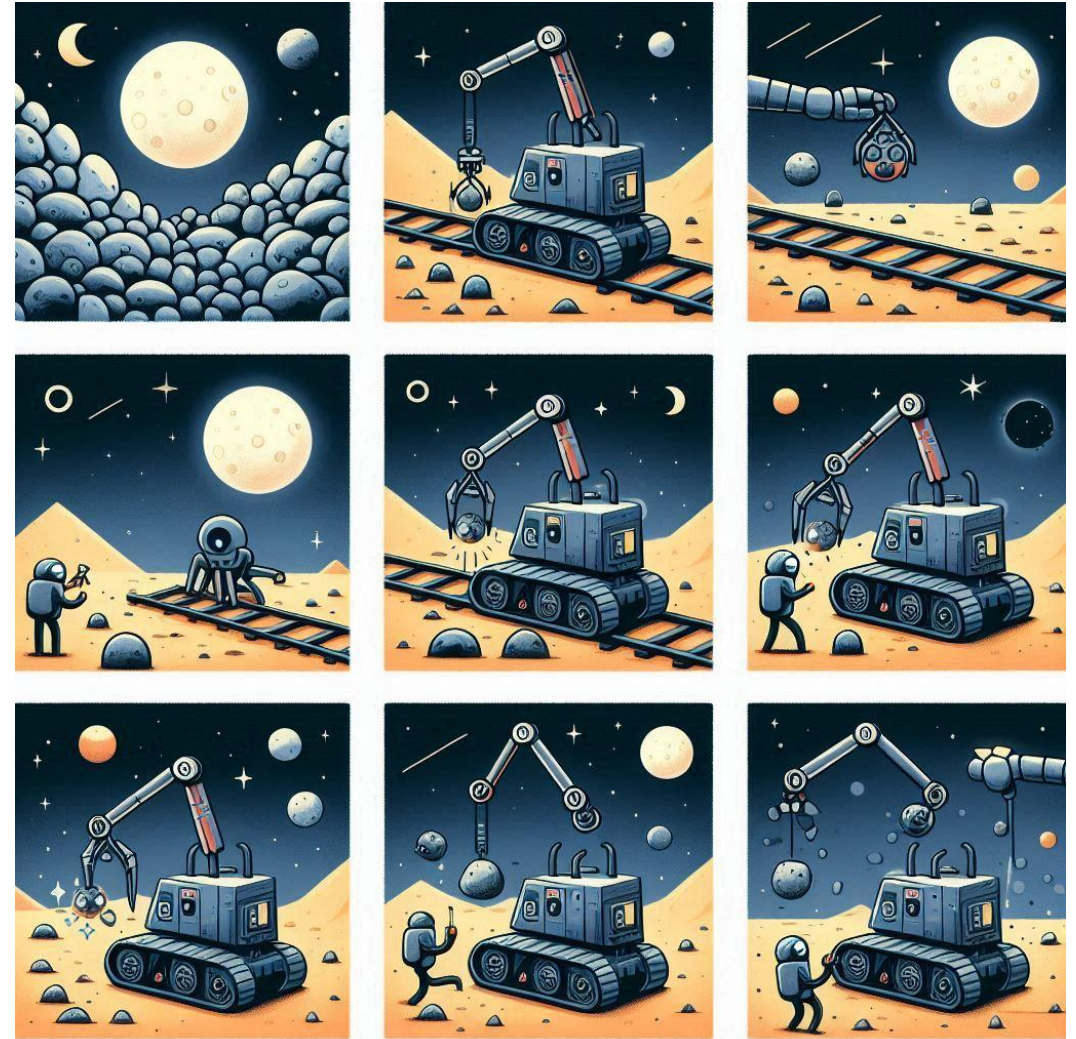
Design Review (Wednesday 30.09)

- 15 Minute Presentation & Demo
- Presentation
 - System Overview
 - ConOps
 - V-Model
 - FMEA
- Demo (voluntary)
 - Present working Baseline Design

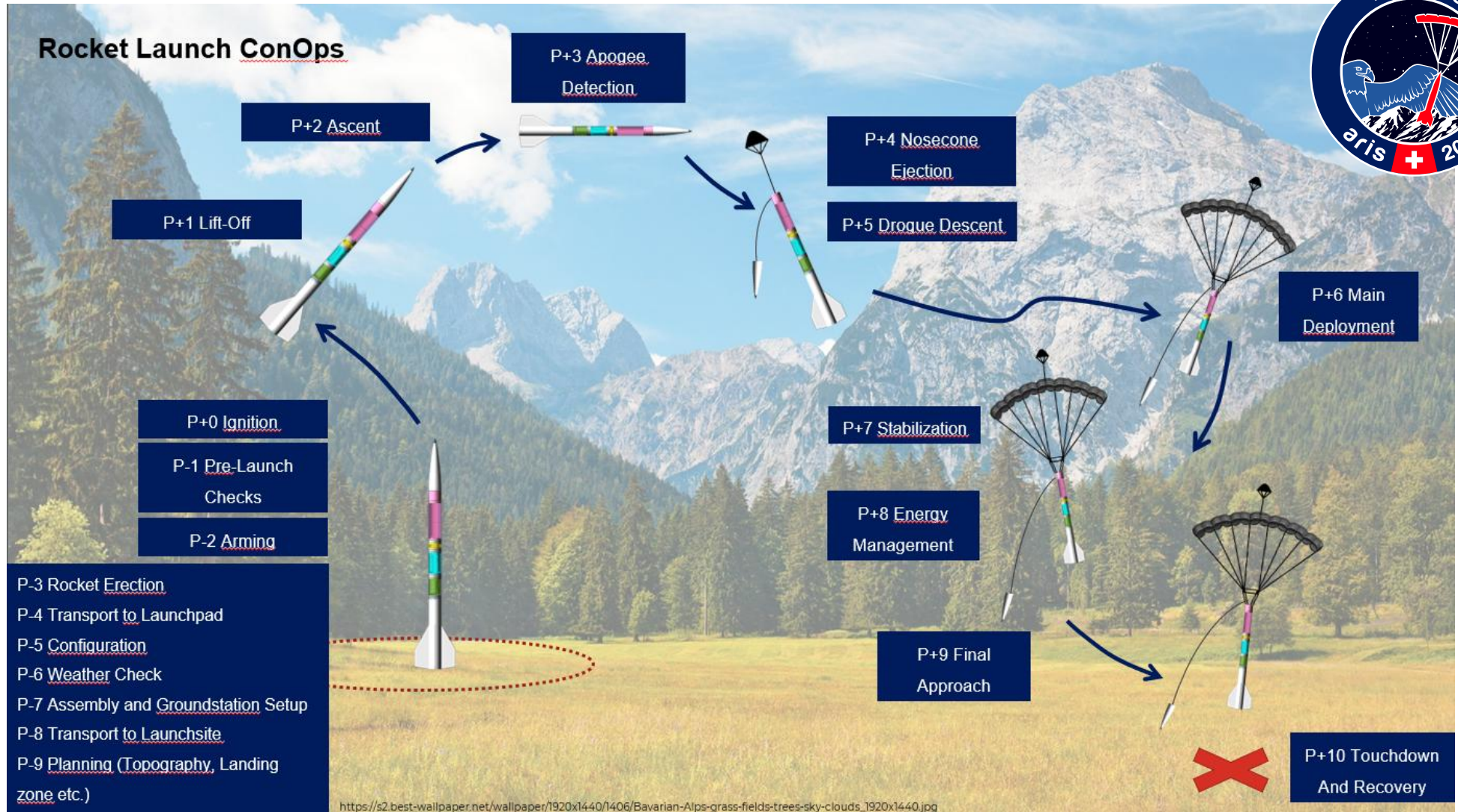


ConOps

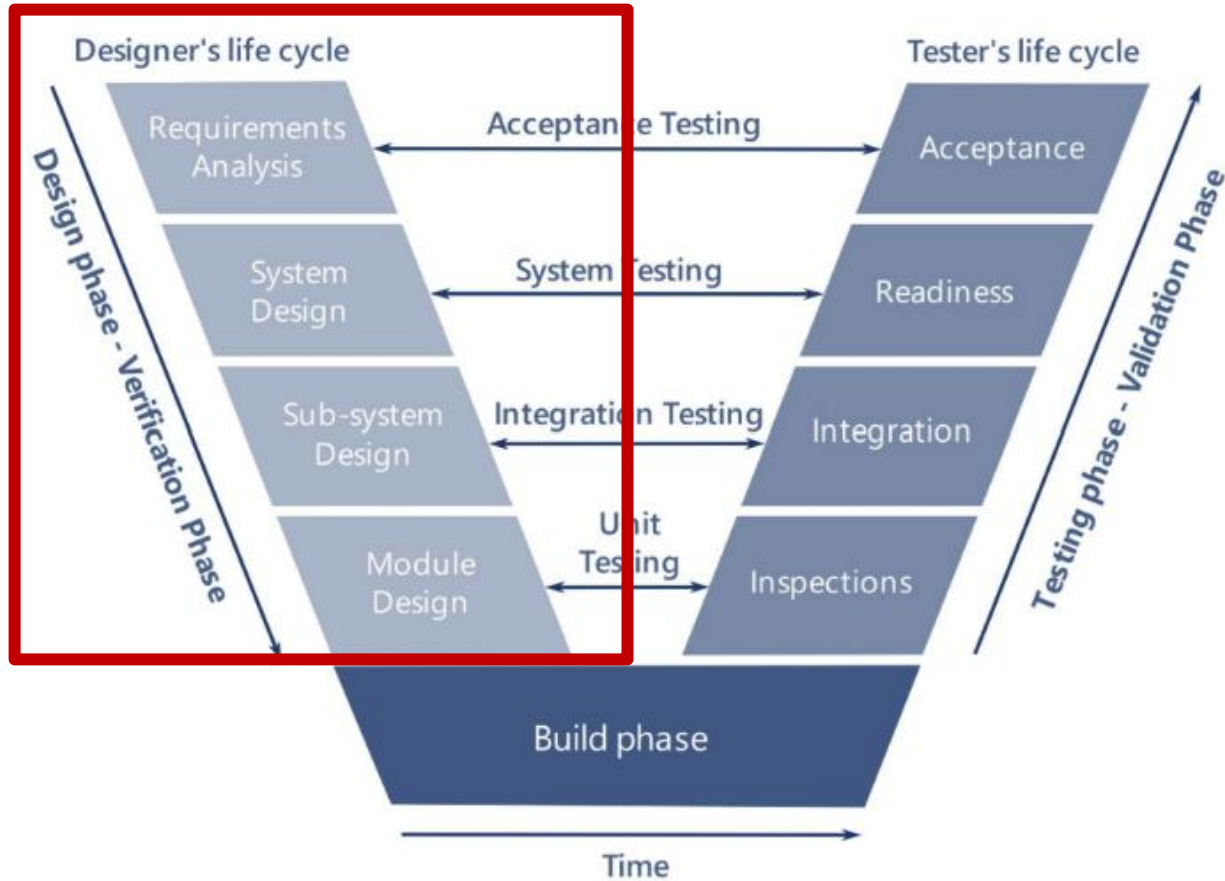
- Describes functionality from a user's perspective
- What does the system do, at what point in time, to complete the mission?
- Big-Picture view



ConOps

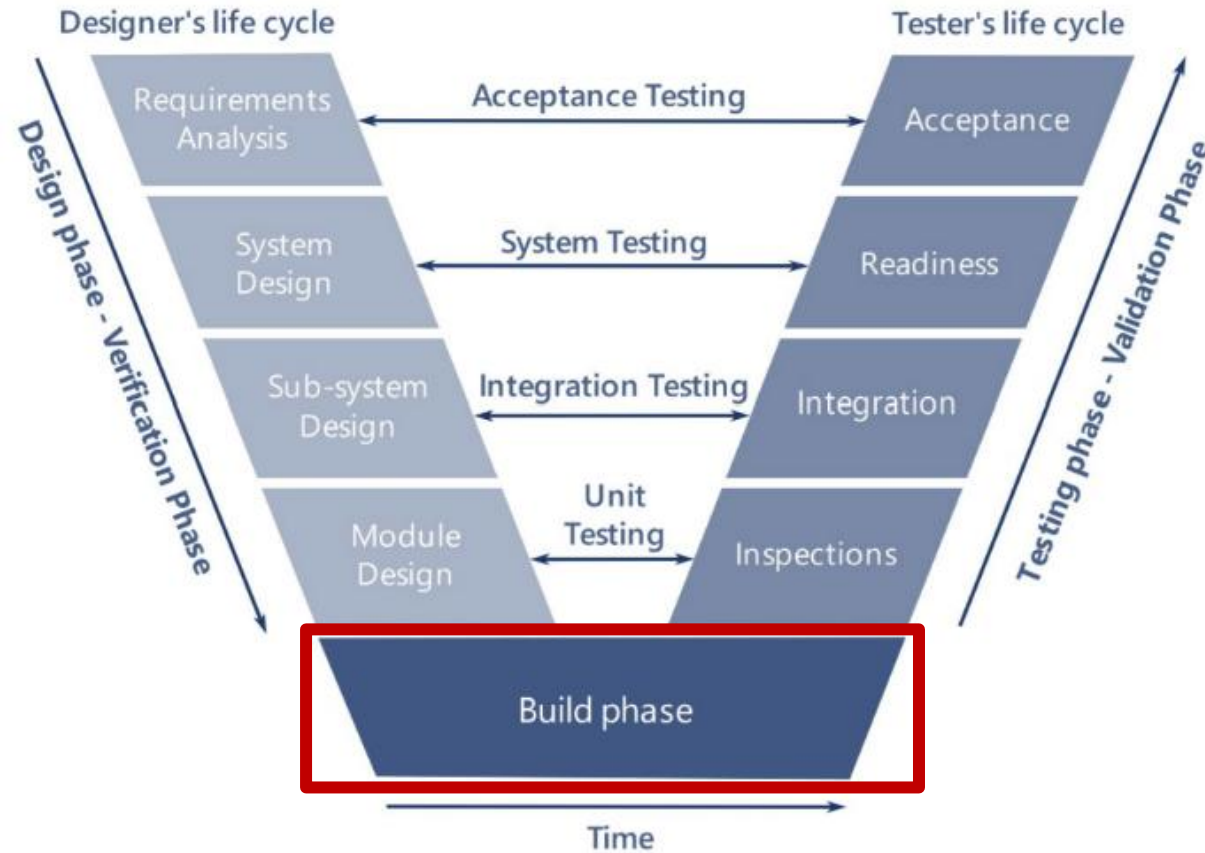


V-Model & FMEA



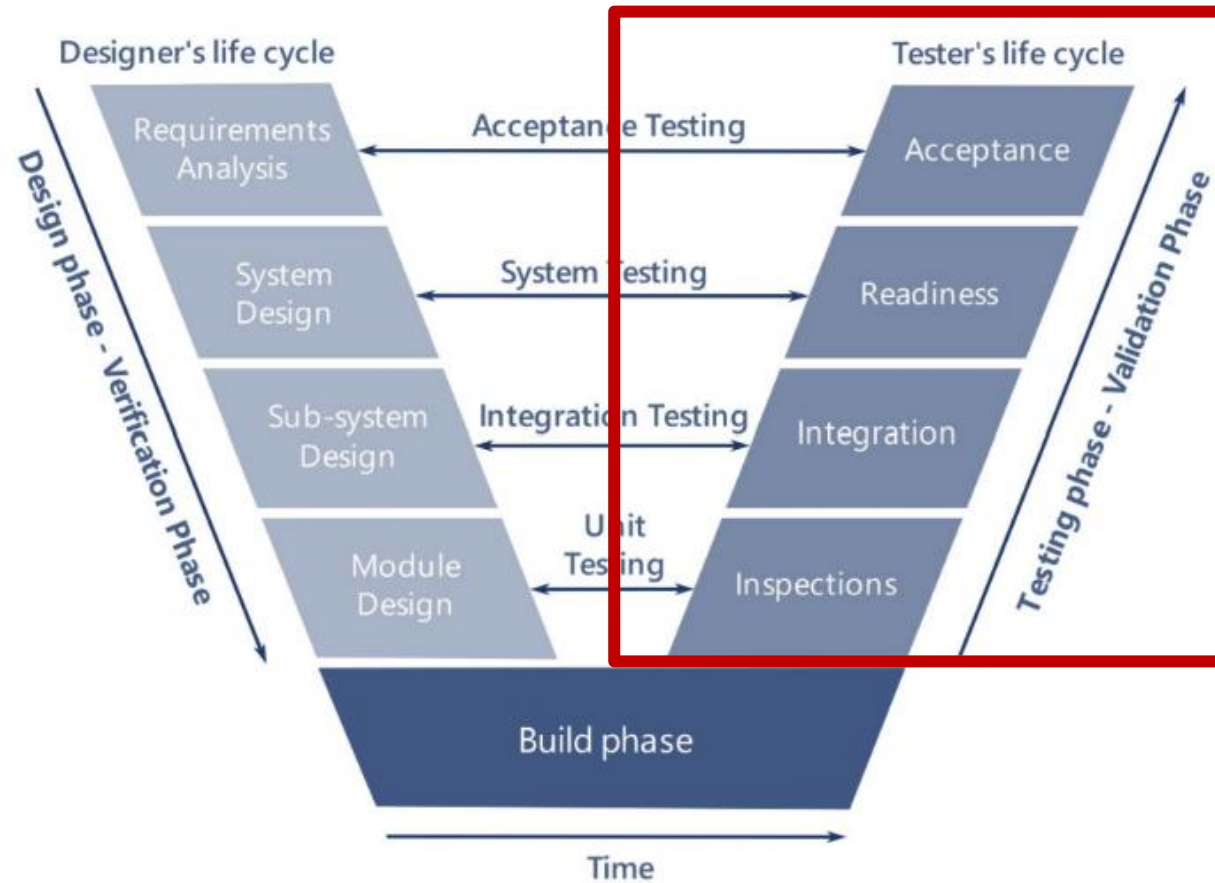
1. Create system break-down

V-Model & FMEA



1. Create system break-down
2. Select **one** sub-system

V-Model & FMEA



1. Create system break-down
2. Select **one** sub-system
3. Conduct FMEA on this sub-system

FMEA – Questions to clarify

Answers to five questions in advance of product implementation/launch:

- What can go wrong?
- Why would that happen?
- What are the consequences?
- How can that be prevented?
- What insights apply to current and future scenarios?

FMEA – Drinking Coffee in a Car

FMEA Identification & definition of the system to be considered Drinking coffee out of a coffee cup in car while driving



						Risk Assessment						
Number	Structure	Function	Potential Failure	Potential Effect	Potential Reason	O	S	D	RPN	Elimination Measure	Assessment	Assignment
1)	Coffee cup	holds coffee	coffee spills out	coffee stains around	breaking	10	4	7	280	fill cup only half	failure accepted	
2)				drivers leg gets burnt	coffee cup is full	6	8	7	336	put cover on top of cup	combine with point 5)	
3)			very hot outside/ burns drivers fingers	driver gets shocked/ lets cup go	did not expect cup being that hot	8	8	6	384	use sleeve on cup	to be implemented	- name of appointee - date to accomplish - validation date
5)	Cup holder	holds cup safely	coffee cup falls in footwell	coffee spills over feet/ leg	driver let cup go when fingers burnt	5	10	8	400	put cover on top of cup	combine with point 7)	
6)	Driving position	focus on traffic	driver gets absent	delay in alerts	tries to grab coffee cup	8	9	9	648	position cup holder in view area of driver	to be implemented	- name of appointee - date to accomplish - validation date
7)					focus on hot spilled coffee	8	9	9	648	see measures to avoid coffee spills out	define effective measure	- name of appointee - date to accomplish - validation date
8)									0	do not drink hot coffee when driving	put warning on dashboard	- name of appointee - date to accomplish - validation date

O = Probability of Occurrence
(failure can happen)

unlikely = 1
slight chance = 2-3
mean = 4-6
moderate = 7-8
high = 9-10

S = Severity
(effect on the environment)

hardly noticeable = 1
negligible effect = 2-3
mean effect = 4-6
grave failure = 7-8
unacceptable failure = 9-10

D = Probability to get Detected
(to avoid severe consequences)

most likely = 1
probable = 2-5
humble = 6-8
poor probability = 9
unlikely = 10

Priority (RPZ)

high RPZ > 399
mean = 50 < RPZ < 399
no = 1 < RPZ < 50

O = Occurrence
S = Severity/Bedeutung
D = Detection
D = Documentation required
RPN = Risk-Priority-Number



https://qsales.qa/cdn/shop/files/DriveSip_Holder_grande.jpg?v=1706011759

Suggested Agenda Today

- 13:15 – 14:00: Introduction & Exercise Overview
 - In Plenum
- 14:00 onwards:
 - Read task description (see Moodle)
 - Work on Data Package
 - Create baseline design
 - Prepare Design Review (Day 2)
- **Hand-Ins** on Moodle
 - DP1: Monday 28. September by 22:59 (TODAY!) -> Feedback Tomorrow Afternoon
 - DR Presentation: Wednesday 30. September by 13:59, Slides as PDF
 - DP2: Thursday 01. October by 22:59

Questions?

One person per team collects Vex Kit from TA's.

Teams

Team 1	Team 2	Team 3	Team 4	Team 5	Team 6	Team 7	Team 8
Joshua D	Illia Y	Leas H	Niam P	Samuel G	Sarah P	Samriddh S	Samuel M
Sebastian N	Fabio G	Fabian Z	Frederik K	Denis N	Héctor F	Lorenzo C	Manuel K
Mikhail K	August L	Loan R	Kevin L	Elia J	Matias B	Gabriel G	Simon R
Traian B	Nina M	Sanjay R	Ciaran M	Tom L	Chrysanthi K	Gina A	Emile S
Mattia D	Till L	Levin S	Manuel M	Aurelio Z	Nicola G	Didier P	Samuel K
David F	Henning W	Lukas G	Christina A	Ryan R	Mariangela M	Alexandra M	

One person per team collects Vex Kit from TA's.